

Chapter 13: Generative AI

Outline

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13.1 Introduction

- ▶ The methods we have studied so far aim to represent new states of a system or process.
- ▶ In this sense, these methods are useful for *understanding* the world around us.
- ▶ But, can we use ML to *imagine* what the world could be?
- ▶ Generative ML is a branch of ML that combines a variety of methods and strategies to mimic creative processes, thus exploring the idea of creating and discovering new systems, from molecules to plants.
- ▶ The combination of methods and applications related to generative ML is what has come to be called *Generative AI*.

13.1 Introduction - Foundation and Importance

- ▶ Generative ML methods are applications that can produce new instances of data based on prior training.
- ▶ As we studied before, ML algorithms are designed to learn the underlying distribution of a dataset or to find the hypothesis that better explains the outcomes of a phenomenon by observing the data available.
- ▶ Generative models take this knowledge and add a new component that uses it to produce new, unseen data points while *maximizing the likelihood that it would belong to the learned distribution or that it would agree with the hypothesis found*.

13.1 Introduction - Foundation and Importance

- ▶ Producing unseen instances of data opens a wide range of applications, such as:
 - ▶ Data augmentation
 - ▶ Data imputation and missing data handling
 - ▶ Creative output generation
 - ▶ Domain adaptation and style transfer
 - ▶ Reinforcement learning by generating diverse and realistic environments.
- ▶ Generative methods in general expand the capabilities of ML applications.

13.3 Self-supervised Learning

- ▶ The first step of any generative ML implementation is learning the data distribution.
- ▶ For most generative ML models, this requires large amounts of *labeled* data which often inaccessible.
- ▶ To deal with this, generative ML uses a different branch of ML: self-supervised learning (SSL).
- ▶ SSL uses a task for which the performance of the model can be verified using the training data itself, without explicit labels.

13.3 Self-supervised Learning

- ▶ Consider, for example, the sequence data discussed in Chapter 9. In a supervised learning setting, we could label sequences as, say, normal or faulty:

$$\langle x_{t=0}, \dots, x_{t=N} \rangle \rightarrow \textit{Faulty} \quad (1)$$

- ▶ In a self-supervised learning setting, on the other hand, we could simply ask the model to predict the next element in the sequence:

$$\langle x_{t=0}, \dots, ? \rangle \rightarrow \langle x_{t=0}, \dots, x_{t=N} \rangle \quad (2)$$

- ▶ With this, we can produce outputs by simply removing information from the inputs. In learning this task, the model extract relevant patterns from the data which can then be leveraged for other tasks.

13.3 Self-supervised Learning

- ▶ Examples of SSL methods include:
 - ▶ Autoencoders
 - ▶ Variational Autoencoders
 - ▶ Generative adversarial networks
 - ▶ Normalizing flows
 - ▶ Autoregressive models

13.4 Variational Autoencoders

- ▶ Autoencoders (AEs) are a class of SSL methods that consists of an encoder and a decoder.
- ▶ The encoder takes an input data sample and maps it to a lower-dimensional *latent space* representation.
- ▶ The decoder takes the latent space representation and reconstructs the original input.
- ▶ By training the AE to minimize the reconstruction error, it learns a compressed representation of the data.
- ▶ By generating new samples in the latent space, we can generate new instances of data using the decoder.
- ▶ In most applications, the encoder and decoder are both ANNs, and can incorporate deep learning architectures.

13.4 Variational Autoencoders

- ▶ Variational autoencoders (VAEs) modify the idea of AEs by learning the probability distribution of the latent space.
- ▶ This improves the likelihood of the generated samples to belong to the actual underlying distribution.

13.4 Variational Autoencoders - Model training

- ▶ During training the encoder and decoder are connected as if they were a single ANN.
- ▶ The parameters $\{\theta_0, \dots, \theta_M\}$ of the distribution that describes the latent space are also learned.

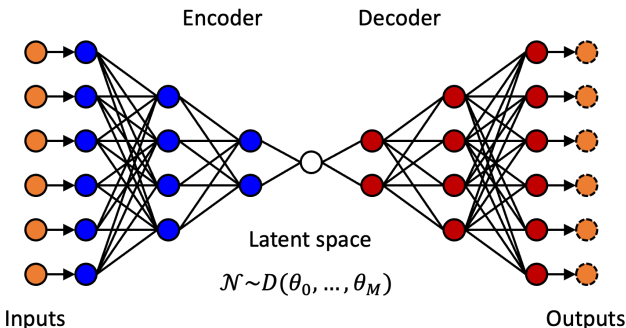


Figure: Autoencoders at training.

13.4 Variational Autoencoders - Generation

- ▶ During inference, only the decoder is used.
- ▶ A sampler produces instances in the latent space following the learned distribution and passes them to the decoder.

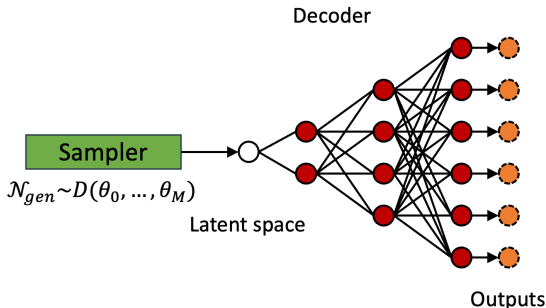


Figure: Autoencoders at generation.

Model explainability and validation

- ▶ While samples generated using the latent space distribution and the decoder are *likely* to belong to the actual distribution this is not guaranteed.
- ▶ Because of this, Generative ML applications in engineering most have verification steps that can corroborate the physical validity of the generated samples.
- ▶ For example, systems that generate new molecules can be paired with mechanistic simulations to evaluate the physical consistency of the molecules generated.
- ▶ Additionally, model explanations can be used for understanding what is being captured by the encoder and how it relates to the physical world.

13.7 Conclusions

- ▶ Generative ML is the combination of ML methods to mimic creative processes.
- ▶ By generating new, unseen samples, generative ML is useful in tasks such as data augmentation, creative output generation, domain adaptation, and others.
- ▶ Self-supervised learning is the cornerstone of generative ML implementations.
- ▶ Explainability and validation remain important steps for the implementation of generative ML in engineering applications.